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IOT-BASED COLD CHAIN MONITORING AND CONTROL SYSTEM FOR PERISHABLE GOODS

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DECLARATION

We hereby declare that we have discussed and agreed to undertake the project titled "**IOT-BASED COLD CHAIN MONITORING AND CONTROL SYSTEM FOR PERISHABLE GOODS**" under the supervision and guidance of **Dr. Eliel Keelson.** We understand the scope of the work and commit to executing the project to the best of our abilities.

Signature: ..... Date: 19/02/2026

Aba Famous Mawulenu Kwaku (20958242)

Signature: ..... Date: 19/02/2026

Ibrahim Ibrahim (20837533)

Signature: ..... Date: 19/02/2026

Asuako Reginald (20908477)

Supervisor: I, **Dr. Eliel Keelson**, hereby confirm my consent to supervise this project and guide the team throughout its lifecycle.

Signature: ..... Date: 19/02/2026

INTRODUCTION

Post-harvest losses remain a significant challenge in Ghana's agricultural sector, particularly in the storage and distribution of perishable commodities such as fruits, vegetables, and fish [2], [4]. Inadequate temperature control, poor storage facilities, and limited cold chain infrastructure contribute to rapid spoilage, resulting in reduced farmer incomes, food insecurity, and economic losses [1], [3]. The absence of proper monitoring systems further worsens the situation, as environmental conditions within storage facilities are often not properly regulated [4].

Advancements in Internet of Things (IoT) technology provide an opportunity to improve cold storage management through real-time monitoring and automated environmental control [5], [6]. This project therefore seeks to design and implement an IoT-based cold chain monitoring and control system capable of regulating temperature and humidity within a storage chamber while providing a software platform for data visualisation and remote supervision. The proposed system aims to enhance storage efficiency and reduce post-harvest losses in Ghana.

PROBLEM STATEMENT

Ghana continues to experience significant post-harvest losses, particularly in the storage and handling of perishable agricultural produce [2], [4]. Inadequate cold chain infrastructure, unreliable power supply, and the absence of proper environmental monitoring systems contribute to rapid deterioration of products before they reach the market [1], [3]. Many smallholder farmers and agribusiness operators lack access to affordable and efficient storage technologies, resulting in reduced product quality, income losses, and increased food insecurity [4].

Furthermore, existing cold storage facilities often operate without real-time monitoring or automated control of temperature and humidity conditions [6]. This makes it difficult to maintain optimal storage environments and respond promptly to fluctuations that may cause spoilage. There is therefore a need for an intelligent, cost-effective system that can monitor and regulate storage conditions while enabling remote supervision through a software platform.

AIM AND OBJECTIVES

Aim:

To design and implement an IoT-based monitoring and control system for cold storage of post-harvest perishable agricultural products.

Specific Objectives:

i. To design and develop a hardware architecture that monitors and controls temperature and humidity in storage facilities

ii. To design and implement a software platform for real-time remote monitoring, data visualization, and control of storage conditions (temperature and humidity).

iii. To test and evaluate the performance of the complete system under simulated storage conditions using standardized metrics.

SIGNIFICANCE OF PROJECT

This project is significant because it provides a technological approach to addressing post-harvest losses in Ghana through improved environmental monitoring and control [2], [4]. By maintaining appropriate temperature and humidity levels within storage facilities, the proposed system can help reduce spoilage of perishable agricultural products, thereby improving food quality, availability, and market value [1], [3]. The solution is particularly relevant to smallholder farmers and agribusiness operators who require affordable and efficient cold storage systems [4].

In addition, the project contributes to the advancement of smart agricultural technologies by integrating embedded systems, IoT communication, and software-based data visualisation within a single platform [5], [6]. The development of a remote monitoring and control system demonstrates the practical application of computer engineering principles to real-world agricultural challenges, supporting national efforts toward food security, economic growth, and sustainable development [3].

METHODOLOGY

The project will adopt a system design and implementation approach consisting of hardware development, software development, and system integration.

1. **Hardware Development**

   The hardware subsystem will comprise the following components:

i. ESP32 microcontroller for data acquisition and Wi-Fi communication
ii. Temperature and humidity sensor (DHT22) for environmental monitoring
iii. Thermoelectric Peltier module for temperature control
iv. MOSFET driver circuit for switching and PWM-based power regulation
v. Humidifier module for humidity control
vi. 12V DC power supply unit

The ESP32 will continuously read sensor data and compare it with predefined threshold values. Based on this comparison, it will automatically regulate the cooling and humidification systems.

2. Software Development

The software subsystem will involve:

i. Programming the ESP32 using Embedded C/C++ in Arduino IDE
ii. Implementing IoT communication using MQTT or HTTP protocol
iii. Developing a cloud-based database for data storage
iv. Designing a web-based dashboard for:
    a. Real-time data visualization
    b. Historical data logging
    c. Remote threshold configuration
    d. Manual system control

3. System Testing and Evaluation

The prototype will be tested under simulated storage conditions to evaluate:

i. Sensor accuracy
ii. Temperature and humidity control stability
iii. System response time
iv. Reliability of wireless data transmission

EXPECTED DELIVERABLES

At the completion of the project, the following deliverables are expected:

i. A functional IoT-based cold chain monitoring and control prototype capable of regulating temperature and humidity within a storage chamber.

ii. A web-based software platform for real-time data visualisation, historical data logging, and remote system control.

iii. A cloud database for secure storage and retrieval of environmental data.

iv. A final project report detailing system design, implementation process, testing procedures, results, and recommendations.

PROJECT TIMELINE

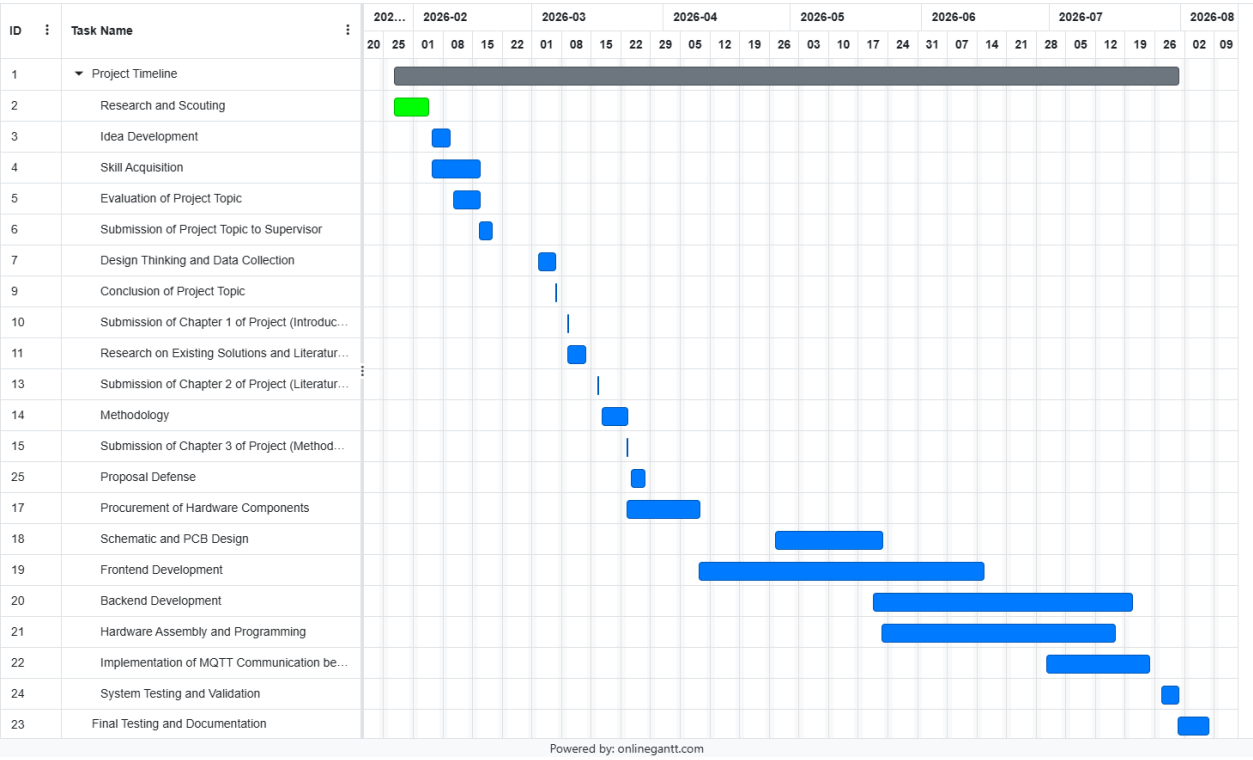

Figure 1: Gantt Chart of Project Timeline

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